$$\text{Now,}\quad \text{Volume of the box} = \text{Volume of the cuboid} + \frac{1}{2}(\text{Volume of cylinder})$$

$$= (l \times b \times h_1) + \frac{1}{2}(\pi r^2 h_2) \text{ cu. units}$$

$$= \left(30 \times 15 \times 10\right) + \frac{1}{2}\left(\frac{22}{7} \times \frac{15}{2} \times \frac{15}{2} \times 30\right)$$

$$= 4500 + 2651.79 = 7151.79$$

Therefore, Volume of the box = 7151.79 $cm^3$

**Example 7.26** Arul has to make arrangements for the accommodation of 150 persons for his family function. For this purpose, he plans to build a tent which is in the shape of cylinder surmounted by a cone. Each person occupies 4 sq. m of the space on ground and 40 cu. meter of air to breathe. What should be the height of the conical part of the tent if the height of cylindrical part is 8 m?

***Solution*** Let $h_1$ and $h_2$ be the height of cylinder and cone respectively.

$$\text{Area for one person} = 4 \text{ sq. m}$$

$$\text{Total number of persons} = 150$$

$$\text{Therefore, total base area} = 150 \times 4$$

$$\pi r^2 = 600 \quad \ldots (1)$$

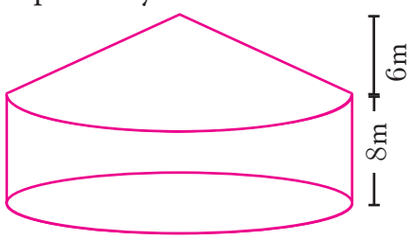


Fig. 7.40

Volume of air required for 1 person = 40 $m^3$

Total Volume of air required for 150 persons = $150 \times 40 = 6000$ $m^3$

$$\pi r^2 h_1 + \frac{1}{3}\pi r^2 h_2 = 6000$$

$$\pi r^2\left(h_1 + \frac{1}{3}h_2\right) = 6000$$

$$600\left(8 + \frac{1}{3}h_2\right) = 6000 \qquad [\text{using } (1)]$$

$$8 + \frac{1}{3}h_2 = \frac{6000}{600}$$

$$\frac{1}{3}h_2 = 10 - 8 = 2$$

$$h_2 = 6 \text{ m}$$

Therefore, the height of the conical tent $h_2$ is 6 m

**Example 7.27** A funnel consists of a frustum of a cone attached to a cylindrical portion 12 cm long attached at the bottom. If the total height be 20 cm, diameter of the cylindrical portion be 12 cm and the diameter of the top of the funnel be 24 cm. Find the outer surface area of the funnel.



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